

Capstone Project - Planning Document

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Project Overview

Website	snitch.com	Framework	Playwright v1.4x+ JavaScript Node.js
Domain	Fashion E-Commerce B2C	Pattern	Page Object Model (POM) + Custom Fixtures
Test Cases	139+ (8 modules x 15 each)	Reporting	Allure Report + Playwright HTML Report
AI Tools	GitHub Copilot + ChatGPT	CI/CD	GitHub Actions

Services & Test Case Breakdown

SR.	Service / Module	Key Test Scenarios	Cases
1	Authentication	Valid login, invalid login, empty credential validation, wrong password validation, signup validation, already registered email validation, logout validation, logged-in navbar validation, login form UI validation, signup form validation, subscription section validation, contact link validation, redirected logged-in user validation, data-driven login validation, browser field validation.	15
2	Product Catalog	Product page visibility, product title validation, product listing validation, product name and price validation, product image validation, product search, no-result search validation, search reset validation, product detail navigation, product category validation, brand filter validation, add-to-cart from product listing, add-to-cart from product detail page, confirmation modal validation, product metadata validation.	15
3	Shopping Cart	Add single product to cart, add multiple products to cart, duplicate product quantity validation, cart row visibility, cart product name validation, cart price validation, cart quantity validation, cart total visibility, remove product validation, empty cart validation, guest cart validation, checkout modal validation, cart navigation validation, product detail navigation from cart, cart item persistence validation.	15
4	Checkout Process	Logged-in checkout flow, checkout page URL validation, delivery address validation, billing address validation, order review table validation, product name validation in order summary, product price validation, quantity validation, total price visibility, comment box validation, place order navigation, guest checkout modal	15

		validation, register/login navigation, order confirmation continue navigation, cart empty validation after order.	
5	Payment	Payment page navigation, payment heading validation, name on card field validation, card number field validation, CVC field validation, expiry month validation, expiry year validation, submit button validation, card input validation, empty payment form validation, missing field validation, checkout-to-payment flow validation, order summary before payment, guest checkout restriction, successful payment continue button validation.	15
6	User Profile	Logged-in username visibility, logout link visibility, signup/login link hidden after login, username text validation, session persistence after navigation, account information heading validation, title radio button validation, signup name prefill validation, signup email prefill validation, password field input validation, date of birth dropdown validation.	15
7	API Testing	Get all products API validation, POST products API validation, get all brands API validation, PUT brands API validation, search product API validation, search without parameter validation, verify login without email validation, DELETE login API validation, invalid login API validation, products response body validation, brands response body validation.	15
8	Home / UI Validation	Homepage load validation, base page accessibility validation, application launch validation, homepage title and page availability check.	15

Implementation Milestones

Milestone	Deliverable
M1 - Setup & Config	Node.js workspace initialized, playwright.config.js configured with cross-browser profiles (Chromium, Firefox, WebKit), package.json scripts configured (test:headed, test:ui, test:report, test:allure).
M2 - Base POM & Locators	Core Page Object classes created (LoginPage, HomePage, ProductPage, CartPage). Robust locators implemented using getByRole(), getByText(), and stable selectors. Test data maintained using JSON files.
M3 - Advanced UI & Actions	Handling dynamic filters, hover interactions on navigation menus, cart table manipulations, responsive layouts, browser dialogs, file uploads, and multi-tab/window handling.
M4 - E2E & Assertions	Linking Page Objects to build complete End-to-End workflows (Login → Search → Add to Cart → Checkout). Implementation of soft assertions for non-critical UI validations.

M5 - Reporting & CI/CD	Integration of Allure Reports with screenshot, trace, and video attachments. Playwright HTML Reports configured. Failure artifact generation enabled. Basic GitHub Actions/Jenkins pipeline configured for headless execution.
M6 - Mocking & Final Polish	Using page.route() to intercept and mock backend API responses for edge-case testing. Final framework refactoring, reusable utility optimization, JSDoc comments, and README documentation completion.

Additional Framework Features

- Cross-browser testing support
- Parallel execution support
- Retry mechanism for flaky tests
- Screenshot capture on failure
- Trace and video recording
- Environment-based configuration handling
- Data-driven testing support
- Reusable utility methods
- Responsive viewport validation
- Structured reporting and debugging support

Website URL

<https://automationexercise.com/>